

Osterquill Toxicology Inc

CVL kinase series: cardiac ion-channel and CYP interaction triage

Final Report

Study OT-20-S0178 (LIMS OQ/2020/002872)

Client: Corvenlea (US)

Study Director: Dr N. Takamori

Report date: October 20, 2020

1 Summary

40 compounds from the Corvenlea aminopyrimidine series 01 were tested for Corvenlea (US) in the hERG (IKr) safety pharmacology, CYP safety interaction panel. Testing started on October 6, 2020 and results were reported on October 20, 2020. All runs met control validity criteria. No compound inhibited hERG with an IC₅₀ below 1 μ M. Most of the set fell in the 1-10 μ M band, so the margin will depend on free therapeutic exposure. Inhibition above 50% at 10 μ M was most frequent for CYP2C9.

Table 1 Study summary

Metric	Value
Project	OT-20-S0178
Client	Corvenlea (US)
Compounds tested	40
Series / chemotype	Corvenlea aminopyrimidine series 01 (Aminopyrimidine)
Assays	hERG patch clamp, CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 μ M
Turnaround (calendar days)	14
Price	USD 22,000
hERG IC ₅₀ <1 μ M	0
hERG IC ₅₀ 1-10 μ M	25
hERG IC ₅₀ $\geq$ 10 μ M	15
hERG median IC ₅₀ (μ M)	8.8
E-4031 control IC ₅₀ (nM)	13
CYP1A2 >50% inhibition @10 μ M	7
CYP2C9 >50% inhibition @10 μ M	13
CYP2C19 >50% inhibition @10 μ M	9
CYP2D6 >50% inhibition @10 μ M	3
CYP3A4 >50% inhibition @10 μ M	11
Compounds with any CYP >50%	18
Ketoconazole (3A4) IC ₅₀ (μ M)	0.030

Compliance statement

This exploratory in vitro investigation is not conducted under GLP; procedures follow Osterquill standard operating procedures and the investigation is audited by Quality Assurance for data integrity only.

2 Objective

To rank the test article cohort for hERG (IKr) safety pharmacology, CYP safety interaction panel liability so that Corvenlea (US) could choose analogues for progression.

3 Test items

40 compounds were received as 10 mM DMSO solutions in one test article cohort on October 6, 2020 and logged against OQ/2020/002872. Identity and concentration were taken as supplied by the client.

4 Methods

hERG (IKr) safety pharmacology

HEK293 cells stably expressing hERG (KCNH2) were recorded in whole-cell configuration on a 16-channel automated patch clamp platform at room temperature. Cells were held at -80 mV, depolarised to +40 mV for 2 s and repolarised to -50 mV for 2 s to elicit the tail current, at 0.1 Hz. Each compound was applied cumulatively at 0.12, 0.37, 1.1, 3.3, 10 and 30 μ M (0.3% DMSO final), with 3-5 minutes per concentration.

- Control validity: seal resistance ≥ 500 M Ω before break-in and ≥ 200 M Ω throughout
- Tail current amplitude ≥ 200 pA at baseline; run-down $\leq 10\%$ over the vehicle period
- Positive control E-4031 on every run; IC50 must fall within 5-50 nM
- IC50 from a four-parameter logistic fit of mean tail-current inhibition; $n \geq 2$ cells per compound

CYP safety interaction panel

Pooled human liver microsomes (0.1 mg/mL protein) were incubated with each compound at 10 μ M and an isoform-selective probe substrate at approximately K_m , in duplicate, for 10 minutes at 37 °C after NADPH addition. Metabolite formation was measured by LC-MS/MS against vehicle (0.1% DMSO) incubations.

Probe substrates and controls

Isoform	Probe substrate	Metabolite	Positive control
CYP1A2	Phenacetin	Acetaminophen	Furafylline
CYP2C9	Diclofenac	4'-Hydroxydiclofenac	Sulfaphenazole
CYP2C19	S-Mephenytoin	4'-Hydroxymephenytoin	Ticlopidine
CYP2D6	Dextromethorphan	Dextrorphan	Quinidine
CYP3A4	Midazolam	1'-Hydroxymidazolam	Ketoconazole

Results are reported as mean percent inhibition relative to vehicle. Values above 50% at 10 μ M are flagged for follow-up; small negative values indicate no inhibition.

5 Results

hERG results

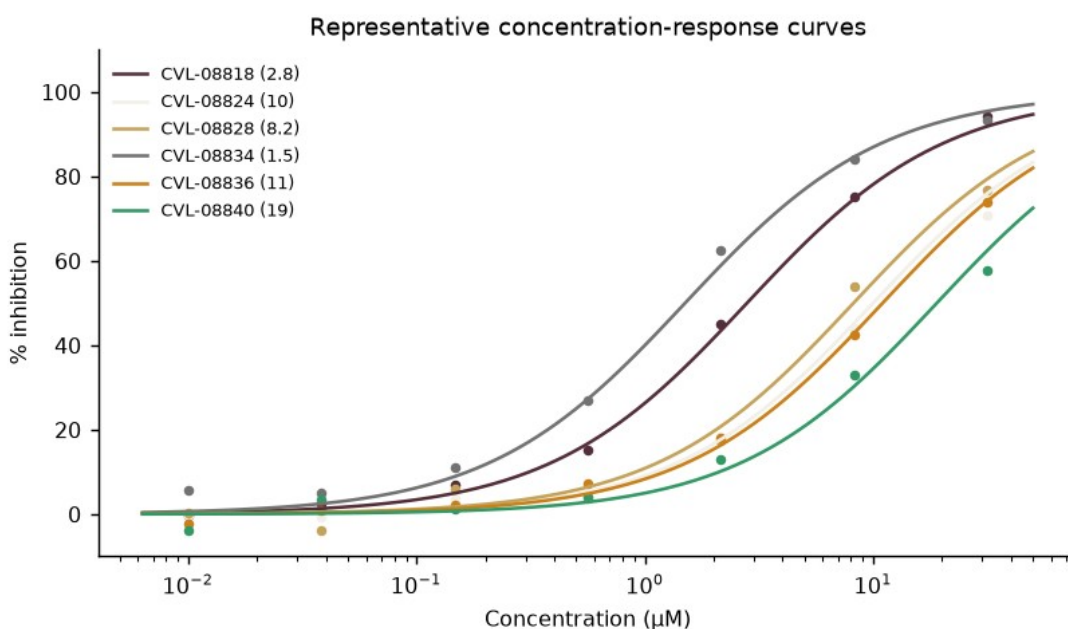
All 40 compounds were tested. No compound inhibited hERG with an IC₅₀ below 1 µM. Most of the set fell in the 1-10 µM band, so the margin will depend on free therapeutic exposure. The median IC₅₀ was 8.8 µM (0 below 1 µM, 25 between 1 and 10 µM, 15 at or above 10 µM). E-4031 gave an IC₅₀ of 13 nM, within the acceptance range.

Table 2 hERG IC₅₀ by compound (µM)

Compound	IC ₅₀ rep 1 (µM)	IC ₅₀ rep 2 (µM)	IC ₅₀ rep 3 (µM)	IC ₅₀ mean (µM)	% inhibition @10 µM	n	QC note
CVL-08810	>30	>30		>30	28	2	
CVL-08818	2.8	3.5	3.6	3.3	72	3	
CVL-08824	10	10	6.4	8.7	60	3	
CVL-08828	8.2	12		9.9	51	2	
CVL-08834	1.5	2.0		1.7	87	2	
CVL-08836	11	13		12	49	2	
CVL-08840	19	11	11	13	43	3	
CVL-08846	4.1	3.1		3.6	77	2	
CVL-08848	9.9	8.4		9.1	55	2	
CVL-08854	27	19		23	28	2	
CVL-08858	7.4	5.3		6.3	69	2	
CVL-08860	25	>30		>30	29	2	
CVL-08866	4.6	8.9		6.4	64	2	
CVL-08870	27	>30		>30	16	2	
CVL-08878	2.4	1.6	2.1	2.0	88	3	
CVL-08882	>30	>30		>30	26	2	Precipitation observed at top concentration
CVL-08884	23	29	20	24	28	3	
CVL-08888	4.3	2.0		3.0	78	2	
CVL-08890	5.0	6.9		5.8	65	2	
CVL-08894	3.7	2.6		3.1	69	2	
CVL-08896	9.3	18		13	43	2	
CVL-08900	23	>30	26	24	22	3	
CVL-08906	7.3	4.6	12	7.3	54	3	
CVL-08912	3.4	1.9		2.5	71	2	Precipitation observed at top concentration
CVL-08920	9.7	8.6		9.1	46	2	
CVL-08924	4.6	2.9		3.6	75	2	
CVL-08926	9.5	8.3		8.9	56	2	
CVL-08932	9.3	21		14	45	2	
CVL-08936	6.3	6.8		6.5	59	2	
CVL-08938	15	17	12	15	37	3	
CVL-08944	1.9	1.9		1.9	80	2	
CVL-08948	5.9	8.7		7.1	64	2	
CVL-08954	13	13		13	45	2	
CVL-08960	7.3	4.8		5.9	58	2	
CVL-08966	10	10		10	47	2	

CVL-08968	5.4	5.8		5.6	63	2	
CVL-08978	1.4	2.1		1.7	84	2	
CVL-08980	7.9	11		9.3	58	2	
CVL-08984	4.9	6.5		5.7	60	2	
CVL-08986	19	14		16	36	2	

Top concentration 30 μ M; 6-point, 3-fold dilution; positive control E-4031 IC₅₀ 13 nM (acceptance 5-50 nM)



CYP inhibition results

Inhibition above 50% at 10 μ M was most frequent for CYP2C9. In total 18 compounds inhibited at least one isoform by more than 50% at 10 μ M (CYP3A4: 11; CYP2D6: 3; CYP2C19: 9; CYP2C9: 13; CYP1A2: 7). Ketoconazole inhibited CYP3A4 with an IC₅₀ of 0.030 μ M.

Table 3 Percent inhibition at 10 μ M

Compound	CYP1A2 (%)	CYP2C9 (%)	CYP2C19 (%)	CYP2D6 (%)	CYP3A4 (%)
CVL-08810	3	24	22	4	53
CVL-08818	20	70	7	4	27
CVL-08824	4	-2	0	8	-6
CVL-08828	6	-6	27	-1	20
CVL-08834	6	75	83	60	40
CVL-08836	7	39	41	4	46
CVL-08840	-11	0	-6	1	5
CVL-08846	33	69	42	77	17
CVL-08848	-2	-2	1	-6	6
CVL-08854	-2	10	-0	-8	15
CVL-08858	17	15	4	-4	-4
CVL-08860	5	3	1	7	2
CVL-08866	60	50	29	6	10
CVL-08870	9	-7	5	-12	5
CVL-08878	10	3	10	5	11

CVL-08882	13	5	62	-8	28
CVL-08884	0	5	-0	-1	12
CVL-08888	5	52	19	-4	25
CVL-08890	30	38	44	22	36
CVL-08894	19	67	20	3	73
CVL-08896	5	-1	1	-4	-4
CVL-08900	8	-6	5	-4	2
CVL-08906	40	11	7	13	45
CVL-08912	7	7	44	0	7
CVL-08920	-5	-6	-4	-4	0
CVL-08924	7	27	15	-2	12
CVL-08926	98	85	80	100	93
CVL-08932	-3	-7	-10	4	-9
CVL-08936	-1	6	2	-5	6
CVL-08938	-0	3	6	6	2
CVL-08944	23	77	78	10	91
CVL-08948	52	50	68	28	91
CVL-08954	11	11	26	0	68
CVL-08960	45	62	51	18	100
CVL-08966	87	77	96	27	90
CVL-08968	14	81	64	9	26
CVL-08978	56	58	62	26	68
CVL-08980	-2	-1	2	3	-2
CVL-08984	58	24	37	28	84
CVL-08986	88	53	25	9	87

Single concentration 10 μ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC₅₀, μ M): furafylline (CYP1A2) 1.9; sulfaphenazole (CYP2C9) 0.35; ticlopidine (CYP2C19) 1.1; quinidine (CYP2D6) 0.049; ketoconazole (CYP3A4) 0.030

6 Discussion

No compound inhibited hERG with an IC₅₀ below 1 μ M. Most of the set fell in the 1-10 μ M band, so the margin will depend on free therapeutic exposure. Inhibition above 50% at 10 μ M was most frequent for CYP2C9. Results are exploratory and are intended for compound selection; they are not a substitute for a definitive GLP safety pharmacology study.

7 Conclusion

The data support ranking within the set. Recommended next step: select analogues without a hERG or CYP signal for further investigation and repeat any single-run outliers.

Name	Role	Signature	Date
Naomi Takamori	Study Director	<i>Takamori</i>	October 20, 2020
Lucia Moretti	GLP Quality Assurance and Archive Lead	<i>Moretti</i>	October 19, 2020